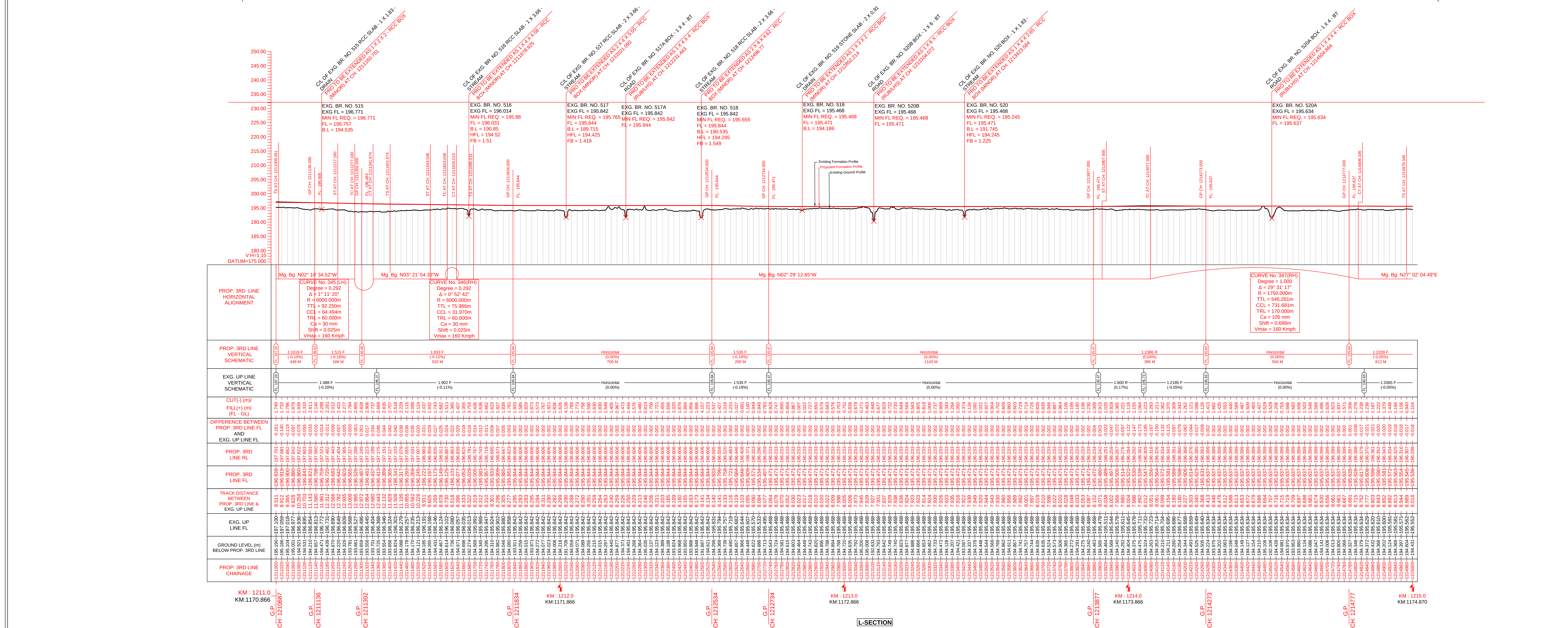
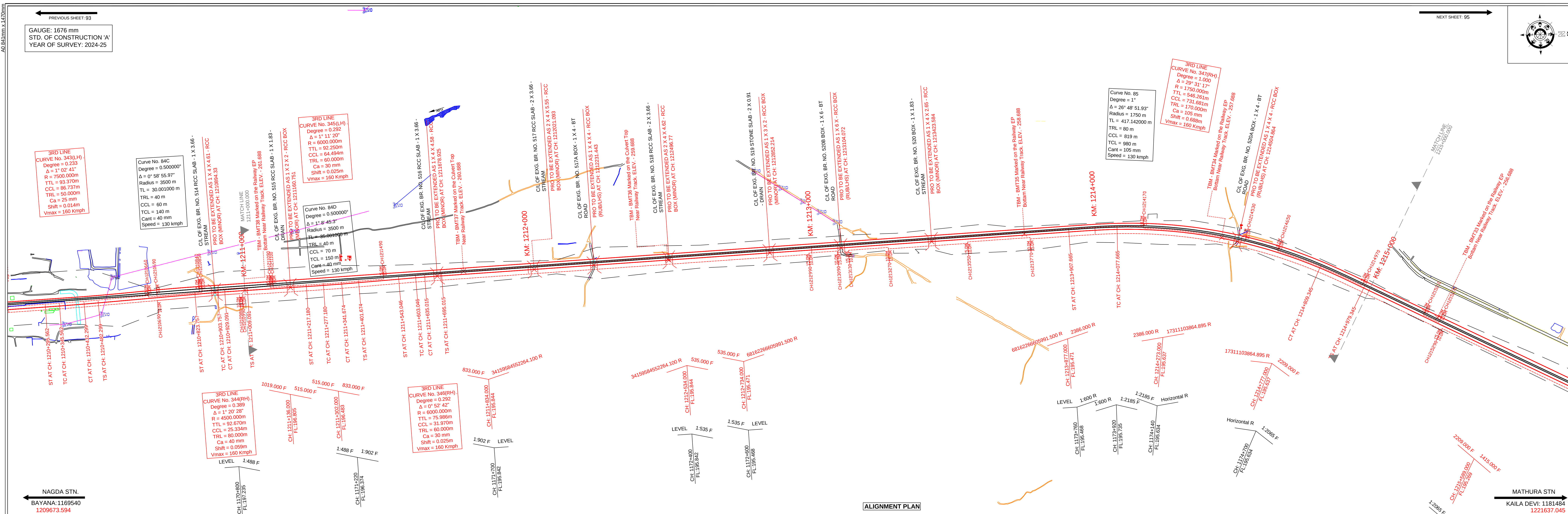




- NOTE:
1. ALL DIMENSIONS ARE IN METERS, UNLESS OTHERWISE SPECIFIED.
 2. EXISTING WORK SHOWN IN BLACK.
 3. PROPOSED WORK SHOWN IN RED.
 4. AS PER TERMS OF REFERENCE YARD GRADIENT HAVE BEEN KEPT TO THE EXTENT POSSIBLE MAINTAINING GRADIENT AS 1 IN 150 (COMPENSATED).
 5. TRACK LEVEL SHOULD BE 762 MM ABOVE FORMATION LEVEL FOR TRACK STRUCTURE OF 60 KG NEW RAIL ON PSC WIDER SLEEPER (1660 NOS PER KM) WITH 350 MM BALLAST CUSHION.
 6. RULING GRADIENT OF THIS SECTION IS 1 IN 150.
 7. AS PER ENGINEERING CODE, VERTICAL CURVE WILL BE PROVIDED AT SITE WHERE ALGEBRAIC DIFFERENCE BETWEEN TWO GRADES IS EQUAL OR MORE THAN 4MM PER METER OR 0.4%.
 8. LAND TO BE ACQUIRED SHOWN IN DASHED RED.
 9. THE CHAINAGE AND KILOMETERS SHOWN ARE RECKONED FROM CENTER LINE OF STATION AS 0+00
 10. THE SECTION IS DESIGNED FOR MAXIMUM SECTIONAL SPEED OF 160 KMPH.
 11. THE PROPOSED FORMATION AND BRIDGES IN THE SECTION IS DESIGNED FOR 25 T-2008 AXLE LOADING.

TBM DETAILS					
TBM ID	Changeage	Easting	Northing	MSL	Description
BMT38	1211000.009	728017.578	2980556.857	261.688	Marked on the Railway EP Bottom Near Railway Track.
BMT37	1211772.337	727970.589	2981327.835	260.688	Marked on the Culvert Top Near Railway Track.
BMT36	1212492.534	727948.075	2982047.713	259.688	Marked on the Culvert Top Near Railway Track.
BMT35	1213750.879	727891.46	2983304.806	258.688	Marked on the Railway EP Bottom Near Railway Track.
BMT34	1214447.372	727922.353	2983998.943	257.688	Marked on the Railway EP Bottom Near Railway Track.

- LEGENDS:**
- | | | | |
|------------------------|---|---------------|---|
| EXISTING RAILWAY TRACK |  | EXISTING ROW |  |
| PROPOSED RAILWAY TRACK |  | METAL ROAD |  |
| PROPOSED ROW |  | UN-METAL ROAD |  |
| PRO. BRIDGES |  | | |
| ELECTRIC LINES |  | | |
| HT LINES |  | | |
| WATER BODIES |  | | |
| RAILWAY STRUCTURES |  | | |

<u>ABBREVIATIONS</u>		
R = Radius	PVI = Point of Vertical Intersection	L = Length
Δ = Angle	TTP1 = Tangent to Transition Point-1	FL = Formation Level
TRL = Transition Length	CTP1 = Curve Tangent Point-1	RL = Rail Level
TL = Tangent Length	CTP2 = Curve Tangent Point-2	G = Gradient
SL = Straight Line Length	TTP2 = Tangent to Transition Point-2	
CL = Curve Length	BVC = Beginning of Vertical Curve	
CCL = Circular Curve Length	EVC = End of Vertical Curve	
IP = Intersection Point	ERL = Existing Rail Level	
RTL = Road Top Level	BL = Bed Level	

REFERENCE DRAWINGS	
DRAWING TITLE	DRAWING No.
DRG NO DY.CE/C(KOTA	
H.Q. DRG. NO. - CPM/KOTA/E	
NAME OF THE OFFICER & DESIGNATION	SIGNATURE
(VIJAY PANDEY) CPM/WCR/KOTA	
(GAURAV MISHRA) DY.CE/C(KOTA	
(RAJESH YADAV) AXEN/KOTA	
(P.K. KANWA) SSE/WCK/KOTA	
(L.S. MEENA) JE/DRG(KOTA	

ISSUE RECORD					
1	01-07-2025	R0	SIVA THUTA	RAMMOHAN	RAMAKRISHNA
No.	DATE	REV	PREPARED BY	CHECKED BY	APPROVED BY

CLIENT:  WEST CENTRAL RAILWAY
KOTA DIVISION

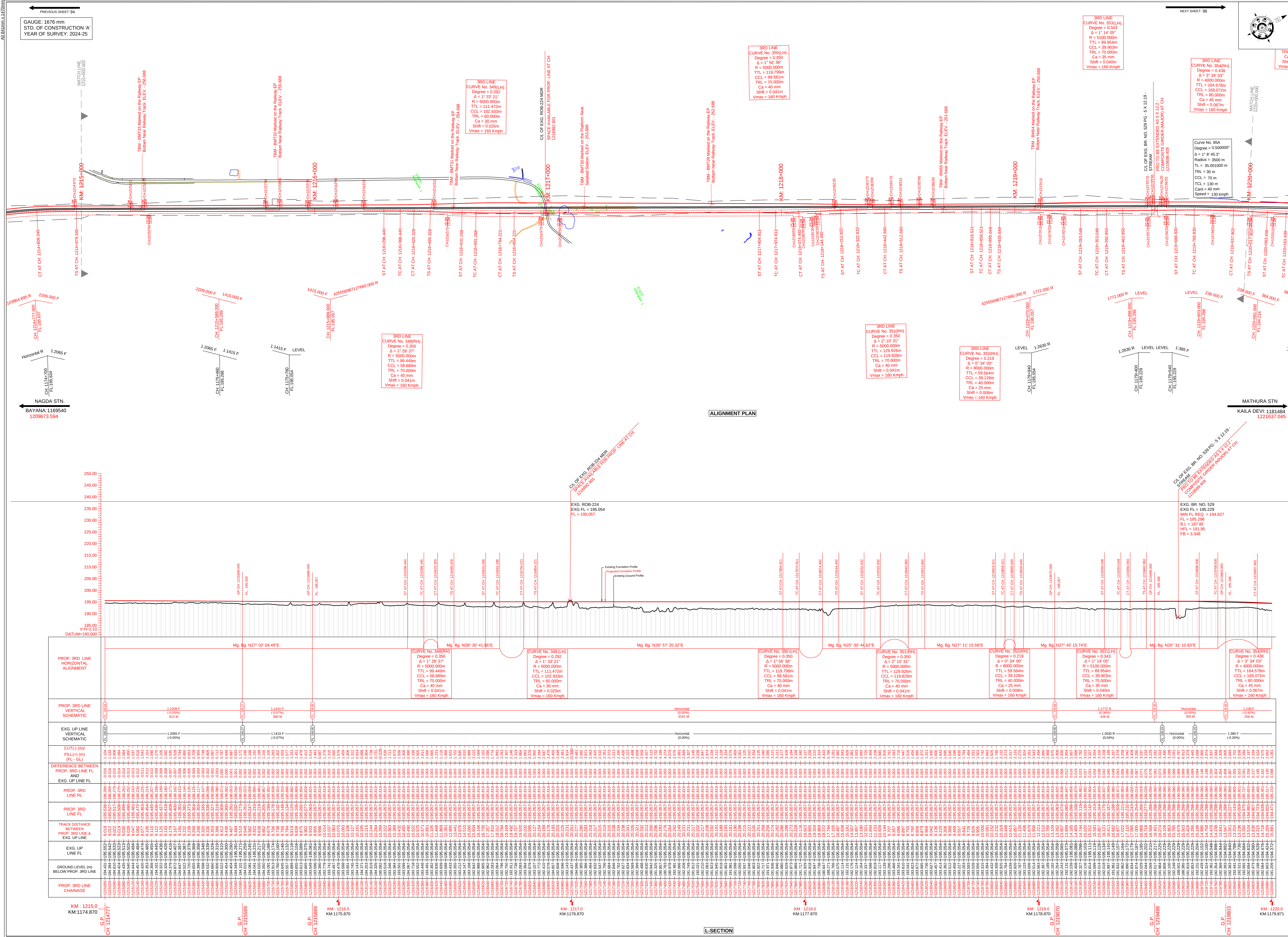
PROJECT : CONDUCTING FINAL LOCATION SURVEY (FLS) INCLUDING PREPARATION OF DPR, GEOTECHNICAL INVESTIGATION, SIGNALING WORK, ELECTRICAL WORKS AND OPERATING/TRAFFIC WORK FOR CONSTRUCTION OF ELECTRIFIED NEW LINE/DOUBLING/THIRD LINE/4TH LINE. FLY OVER WORK/BYPASS LINE ETC.

TITLE : DETAILED PLAN AND L SECTION MATHURA-NAGDA -3RD LINE
1211+000.000 TO 1215+000.000

CONSULTANT :

 **aarvee**
aarvee engineering solutions limited.
(Formerly) Fellow Associates Architects Engineers & Consultants Private Limited.
Registered Office Address: 9-5 Ravia Residency, Strategy Colony Main Road Hyderabad-500002 Telangana, India.
Tel: +91-40-4000040, 23707033; Fax: +91-40-3270777; Email: info@aarvee.com; www.aarvee.com
CIN U47090TG2009PLC001691

PRG.No: AA/WCR/FLS/2616/MKN-3RD/P&P/94	SHEET NO : 94
SCALE : HOR : 1:5000 , VER : 1:500	DATE : 01-07-2025



NOTE:

- ALL DIMENSIONS ARE IN METERS, UNLESS OTHER WISE SPECIFIED.
- EXISTING WORK SHOWN IN BLACK.
- PROPOSED WORK SHOWN IN RED.
- AS PER TERMS OF REFERENCE YARD GRADIENT HAVE BEEN EASED TO THE EXTENT POSSIBLE MAINTAINING GRADIENT AS 1 IN 150 (COMPENSATED).
- RAIL LEVEL SHOULD BE 762 MM ABOVE FORMATION LEVEL FOR TRACK STRUCTURE OF 60 KG NEW RAIL ON PSC WIDER SLEEPER (1660 NOS PER KM) WITH 350 MM BALLAST CUSHION.
- RULING GRADIENT OF THIS SECTION IS 1 IN 150.
- AS PER ENGINEERING CODE, VERTICAL CURVE WILL BE PROVIDED AT SITE WHERE ALGEBRAIC DIFFERENCE BETWEEN TWO GRADES IS EQUAL OR MORE THAN 4MM PER METER OR 0.4%.
- THE CHAINAGE AND KILOMETERS SHOWN ARE RECKONED FROM CENTER LINE OF STATION BUILDING AS 0.00
- THE SECTION IS DESIGNED FOR MAXIMUM SECTIONAL SPEED OF 160 KMPH.
- THE PROPOSED FORMATION AND BRIDGES IN THE SECTION IS DESIGNED FOR 25 T-2008 AXLE LOADING.

TBM DETAILS					
TBM ID	Chainage	Easting	Northing	MSL	Description
BMT33	1215528.131	728224.346	2984750.445	256.688	Marked on the Railway EP Bottom Near Railway Track
BMT32	1215981.008	728537.755	2985402.07	255.688	Marked on the Railway EP Bottom Near Railway Track
BMT31	1216594.565	728830.923	2985941.933	254.688	Marked on the Railway EP Bottom Near Railway Track
BMT30	1217156.715	729087.261	2986441.448	253.688	Marked on the Railway EP Bottom Near Railway Track
BMT29	1217699.261	729323.119	2986930.176	252.688	Marked on the Railway EP Bottom Near Railway Track
BMS5	1218695.916	729798.02	2987806.613	251.688	Marked on the Railway EP Bottom Near Railway Track
BMS4	1219088.967	729967.362	2988161.672	250.688	Marked on the Railway EP Bottom Near Railway Track

LEGENDS:

EXISTING RAILWAY TRACK

PROPOSED RAILWAY TRACK

PROPOSED ROW

PRO. BRIDGES

ELECTRIC LINES

HT LINES

WATER BODIES

RAILWAY STRUCTURES

EXISTING ROW

METAL ROAD

UN-METAL ROAD

ABBREVIATIONS

R = Radius

Δ = Angle

TRL = Transition Length

TL = Tangent Length

SL = Straight Line Length

CL = Curve Length

CCL = Circular Curve Length

IP = Intersection Point

RTL = Road Top Level

PVI = Point of Vertical Intersection

TTT1 = Tangent to Transition Point-1

CTP1 = Curve Tangent Point-1

CTP2 = Curve Tangent Point-2

TTT2 = Tangent to Transition Point-2

BVC = Beginning of Vertical Curve

EVC = End of Vertical Curve

ERL = Existing Rail Level

BL = Bed Level

L = Length

FL = Formation Level

RL = Rail Level

G = Gradient

REFERENCE DRAWINGS	
DRAWING TITLE	DRAWING No.
DRG.No.DY.CE/C(K)OTA	
H.Q. DRG. NO. - CPM/KOTATE	

NAME OF THE OFFICER & DESIGNATION		SIGNATURE	
(VIJAY PANDEY) CPMWC/R(K)OTA			
(GAURAV MISHRA) DY.CE/C(K)OTA			
(RAJESH YADAV) AJEN/KOTA			
(P.K. KANWAR) SSEW/K(K)OTA			
(L.S. MEENA) JE/DRG/KOTA			

ISSUE RECORD				
No.	DATE	REV	PREPARED BY	CHECKED BY
1	01-07-2025	R0	SIVA THUTHA	RAMMOHAN
				RAMAKRISHNA

CLIENT:

WEST CENTRAL RAILWAY
KOTA DIVISION

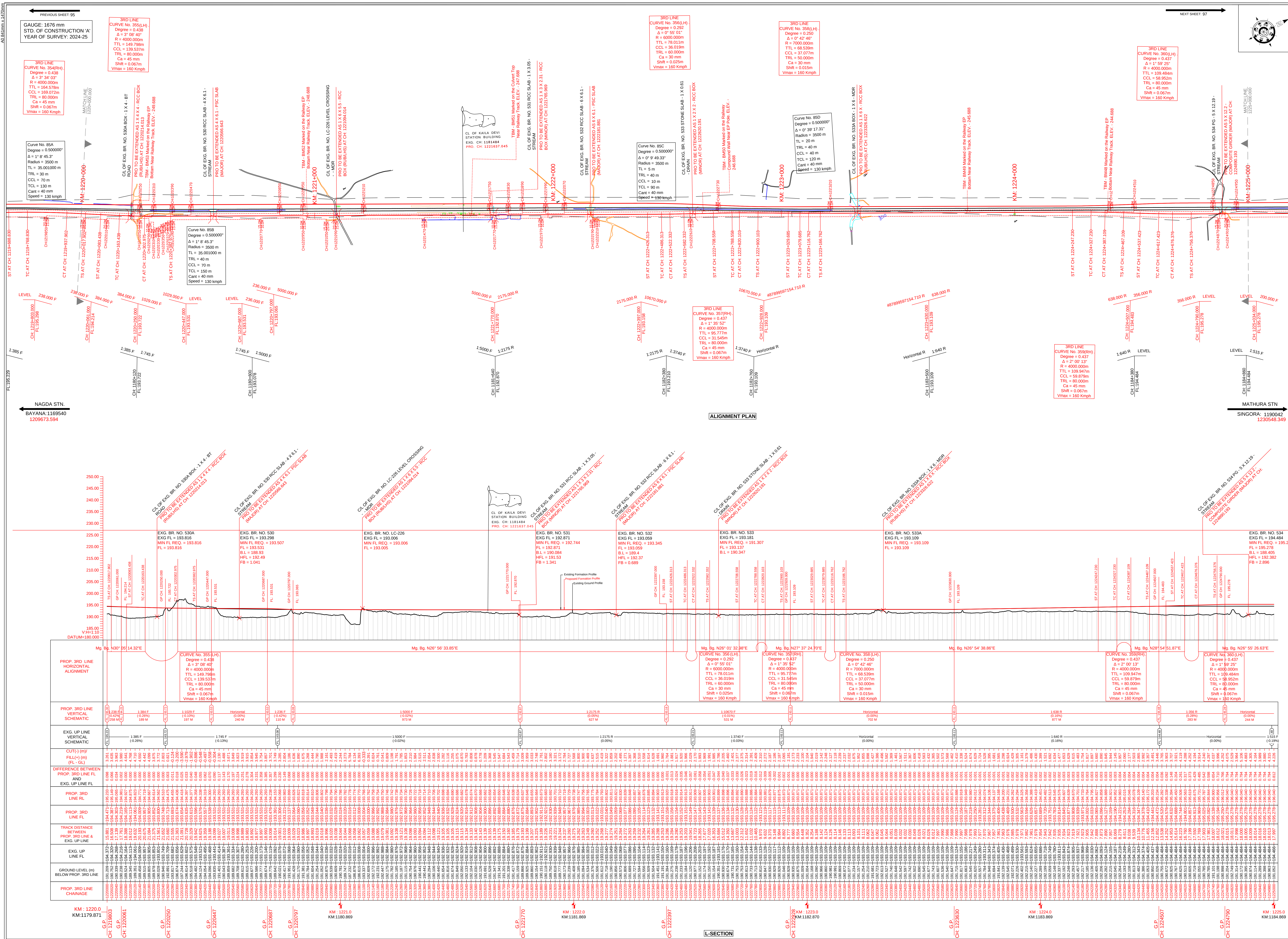
PROJECT : CONDUCTING FINAL LOCATION SURVEY (FLS) INCLUDING PREPARATION OF DPR, GEOTECHNICAL INVESTIGATION, SIGNALING WORK, ELECTRICAL WORKS AND OPERATING/TRAFFIC WORK FOR CONSTRUCTION OF ELECTRIFIED NEW LINE/DOUBLING/THIRD LINE/4TH LINE. FLY OVER WORK/BYPASS LINE ETC.

TITLE : DETAILED PLAN AND L SECTION MATHURA-NAGDA -3RD LINE
1215+000.000 TO 1220+000.000

CONSULTANT :

aarvee engineering consultants limited.

Registered Office Address: A-2, Phase II, Panchsheel Enclave, Mayapuri, New Delhi-110029 (INDIA)
Tel: 011-26108200, 26108201, 26108202, 26108203, 26108204, 26108205, 26108206, 26108207, 26108208, 26108209, 26108210, 26108211, 26108212, 26108213, 26108214, 26108215, 26108216, 26108217, 26108218, 26108219, 26108220, 26108221, 26108222, 26108223, 26108224, 26108225, 26108226, 26108227, 26108228, 26108229, 26108230, 26108231, 26108232, 26108233, 26108234, 26108235, 26108236, 26108237, 26108238, 26108239, 26108240, 26108241, 26108242, 26108243, 26108244, 26108245, 26108246, 26108247, 26108248, 26108249, 26108250, 26108251, 26108252, 26108253, 26108254, 26108255, 26108256, 26108257, 26108258, 26108259, 26108260, 26108261, 26108262, 26108263, 26108264, 26108265, 26108266, 26108267, 26108268, 26108269, 26108270, 26108271, 26108272, 26108273, 26108274, 26108275, 26108276, 26108277, 26108278, 26108279, 26108280, 26108281, 26108282, 26108283, 26108284, 26108285, 26108286, 26108287, 26108288, 26108289, 26108290, 26108291, 26108292, 26108293, 26108294, 26108295, 26108296, 26108297, 26108298, 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26108999, 26109000, 26109001, 26109002, 26109003, 26109004, 26109005, 26109006, 26109007, 26109008, 26109009, 26109010, 26109011, 26109012, 26109013, 26109014, 26109015, 26109016, 26109017, 26109018, 26109019, 26109020, 26109021, 26109022, 26109023, 26109024, 26109025, 26109026, 26109027, 26109028, 26109029, 26109030, 26109031, 26109032, 26109033, 26109034, 26109035, 26109036, 26109037, 26109038, 26109039, 26109040, 26109041, 26109042, 26109043, 26109044, 26109045, 26109046, 26109047, 26109048, 26109049, 26109050, 26109051, 26109052, 26109053, 26109054, 26109055, 26109056, 26109057, 26109058, 26109059, 26109060, 26109061, 26109062, 26109063, 26109064, 26109065, 26109066, 26109067, 26109068, 26109069, 26109070, 26109071, 26109072, 26109073, 26109074, 26109075, 26109076, 26109077, 26109078, 26109079, 26109080, 26109081, 26109082, 26109083, 26109084, 26109085, 26109086, 26109087, 26109088, 26109089, 26109090, 26109091, 26109092, 26109093, 26109094, 26109095, 26109096, 26109097, 26109098, 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- NOTE:
1. ALL DIMENSIONS ARE IN METERS, UNLESS OTHERWISE SPECIFIED
 2. EXISTING WORK SHOWN IN BLACK
 3. PROPOSED WORK SHOWN IN RED.
 4. AS PER TERMS OF REFERENCE YARD GRADIENT HAVE BEEN EASED TO THE EXTENT POSSIBLE MAINTAINING GRADIENT AS 1 IN 10 (COMPENSATED).
 5. RAIL LEVEL SHOULD BE 762 MM ABOVE FORMATION LEVEL FOR TRACK STRUCTURE OF 76 KG NEW RAIL ON PSC WIDER SLEEPER (1560 NOS PER KM) WITH 35% BALLAST CUSHION.
 6. RULING GRADIENT OF THIS SECTION IS 1 IN 150.
 7. AS PER ENGINEERING CODE, VERTICAL CURVE WILL BE PERFORMED AT SITE WHERE ALGEBRAIC DIFFERENCE BETWEEN TWO GRADES IS EQUAL OR MORE THAN 4MM PER METER OR 0.4%.
 8. LAND TO BE ACQUIRED SHOWN IN DASHED RED.
 9. THE CHAINAGE AND KILOMETERS SHOWN ARE RECKONED FROM CENTER LINE OF TRACK AND ARE AS 0.00.
 10. THE SECTION IS DESIGNED FOR MAXIMUM SECTIONAL SPEED OF 160 KMPH.
 11. THE PROPOSED FORMATION AND BRIDGES IN THE SECTION IS DESIGNED FOR 25 T-2008 AXLE LOADING.

TBM DETAILS					
TBM ID	Chainage	Easting	Northing	MSL	Description
BM53	1220220.949	730494.568	2889163.371	249.688	Marked on the Railway EP Bottom Near Railway Track.
BM52	1220967.912	730853.241	2889818.351	248.688	Marked on the Railway EP Bottom Near Railway Track.
BM51	1221671.466	731163.36	2990449.944	247.688	Marked on the Current Top Near Railway Track.
BM50	1222664.793	731804.976	2991339.498	246.688	Marked on the Railway Compound Wall Near EP Pole.
BM49	1223797.219	732122.659	2992364.881	245.688	Marked on the Railway EP Bottom Near Railway Track.
BM48	1224998.008	732299.727	2992860.133	244.688	Marked on the Railway EP Bottom Near Railway Track.

LEGENDS:

EXISTING RAILWAY TRACK		EXISTING ROW	
PROPOSED RAILWAY TRACK		METAL ROAD	
PROPOSED ROW		UN-METAL ROAD	
PRO. BRIDGES			
ELECTRIC LINES			
HT LINES			
WATER BODIES			
RAILWAY STRUCTURES			

ABBREVIATIONS

R = Radius	PVI = Point of Vertical Intersection	L = Length
Δ = Angle	TP1 = Tangent to Transition Point-1	FL = Formation Level
TRL = Transition Length	CTP1 = Curve Tangent Point-1	RL = Rail Level
TL = Tangent Length	CTP2 = Curve Tangent Point-2	G = Gradient
SL = Straight Line Length	TP2 = Tangent to Transition Point-2	
CL = Curve Length	BVC = Beginning of Vertical Curve	
CCL = Circular Curve Length	EVC = End of Vertical Curve	
IP = Intersection Point	ERL = Existing Rail Level	
RTL = Road Top Level	BL = Bed Level	

REFERENCE DRAWINGS

DRAWING TITLE	DRAWING No.
DRG.NO.DY.CE/C/KOTA	
H.Q. DRG. NO. - CPM/KOTA/E	
NAME OF THE OFFICER & DESIGNATION	SIGNATURE
(VIJAY PANDEY) CPM/WCR/KOTA	
(GAURAV MISHRA) DY.CE/C/KOTA	
(RAJESH YADAV) AXEN/KOTA	
(P.K. KANWA) SSE/W/C/KOTA	
(L.S. MEENA) TE/IND/C/KOTA	

[illegible]

1	01-07-2025	R0	SIVA THUTA	RAMMOHAN	RAMAKRISHNA
No.	DATE	REV	PREPARED BY	CHECKED BY	APPROVED BY

LIENT:

WEST CENTRAL RA
KOTA DIVISION

PROJECT: CONDUCTING FINAL LOCATION SURVEY (FLS) INCLUDING PREPARATION OF DPR, GEOTECHNICAL INVESTIGATION, SIGNALING WORK, ELECTRICAL WORKS AND OPERATING/TRAFFIC WORK FOR CONSTRUCTION OF ELECTRIFIED NEW LINE/DOUBLING/THIRD LINE/4TH LINE. FLY OVER WORK/BYPASS LINE ETC.

TITLE : DETAILED PLAN AND L SECTION MATHURA-NAGDA -3RD LINE
1220+000.000 TO 1225+000.000

CONSULTANT :

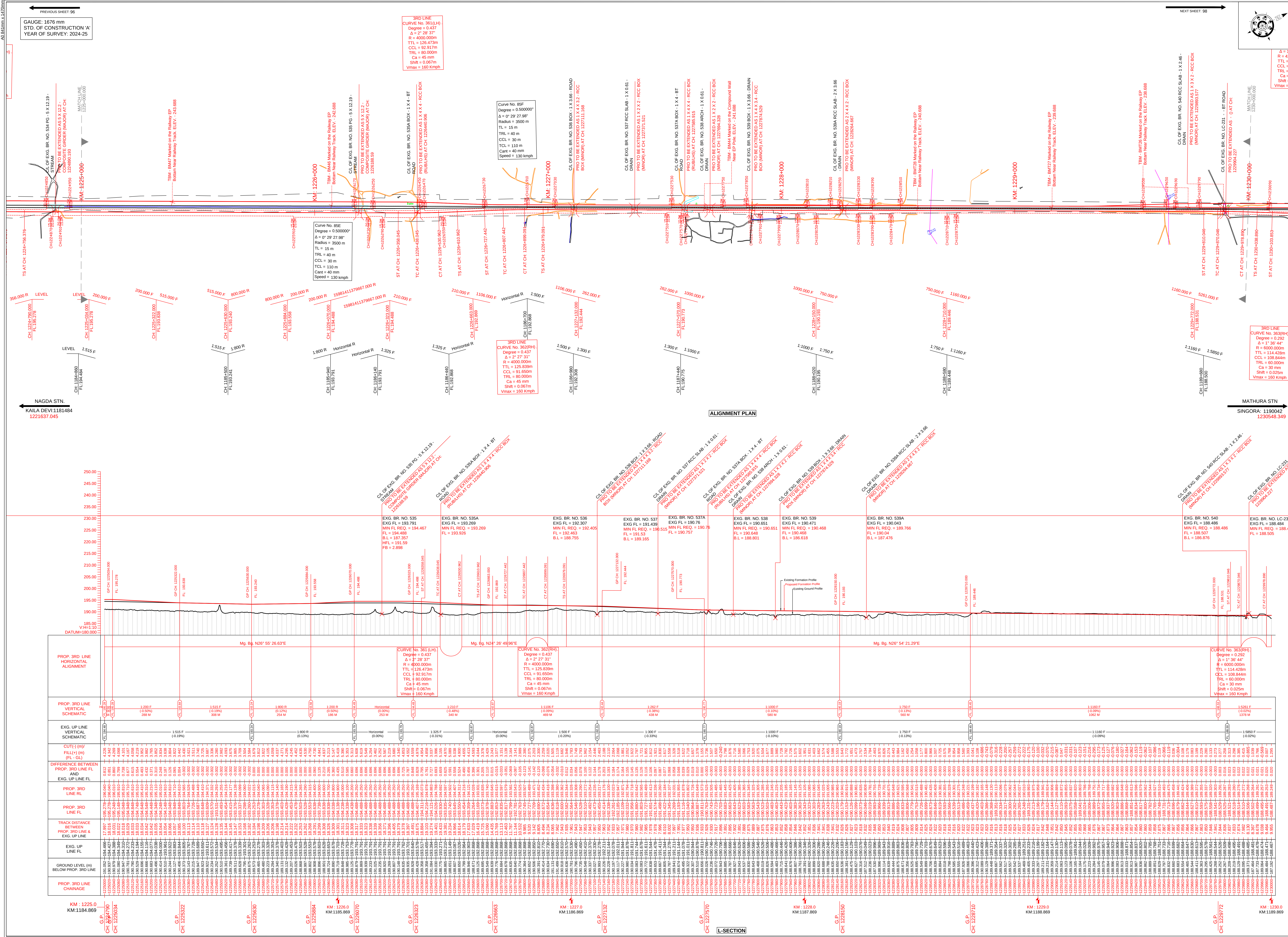


RG.No: AA/WCR/FLS/2616/MKN-3RD/P&P/96

SHEET NO : 96

SCALE : HOB : 1:5000 VER : 1:500

DATE : 01-07-2025



NOTE:

- ALL DIMENSIONS ARE IN METERS, UNLESS OTHER WISE SPECIFIED.
- EXISTING WORK SHOWN IN BLACK.
- PROPOSED WORK SHOWN IN RED.
- AS PER TERMS OF REFERENCE YARD GRADIENT HAVE BEEN EASED TO THE EXTENT POSSIBLE MAINTAINING GRADIENT AS 1 IN 150 (COMPENSATED).
- RAIL LEVEL SHOULD BE 762 MM ABOVE FORMATION LEVEL FOR TRACK STRUCTURE OF 60 KG NEW RAIL ON PSC WIDER SLEEPER (1660 NOS PER KM) WITH 350 MM BALLAST CUSHION.
- RULING GRADIENT OF THIS SECTION IS 1 IN 150.
- AS PER ENGINEERING CODE, VERTICAL CURVE WILL BE PROVIDED AT SITE WHERE ALGEBRAIC DIFFERENCE BETWEEN TWO GRADES IS EQUAL OR MORE THAN 4MM PER METER OR 0.4%.
- LAND TO BE ACQUIRED SHOWN IN DASHED RED.
- THE CHAINAGE AND KILOMETERS SHOWN ARE RECKONED FROM CENTER LINE OF STATION BUILDING AS 0.00
- THE SECTION IS DESIGNED FOR MAXIMUM SECTIONAL SPEED OF 160 KMPH.
- THE PROPOSED FORMATION AND BRIDGES IN THE SECTION IS DESIGNED FOR 25 T-2008 AXLE LOADING.

TBM DETAILS				
TBM ID	Chainage	Easting	Northing	MSL
BM47	122538.752	732823.251	2993777.06	243.688
BM46	1226072.9	733142.604	2994361.072	242.688
BM44	1227669.342	733872.523	2995800.575	241.688
BMT28	1228583.678	734282.535	2996617.838	240.688
BMT27	1229159.958	734540.488	2997133.172	239.688
BMT26	1229742.433	734801.092	2997654.107	238.688

LEGENDS:

EXISTING RAILWAY TRACK

PROPOSED RAILWAY TRACK

PROPOSED ROW

PRO. BRIDGES

ELECTRIC LINES

HT LINES

WATER BODIES

RAILWAY STRUCTURES

EXISTING ROW

METAL ROAD

UN-METAL ROAD

ABBREVIATIONS

R = Radius

Δ = Angle

TRL = Transition Length

TL = Tangent Length

SL = Straight Line Length

CL = Curve Length

ICL = Circular Curve Length

PVI = Point of Vertical Intersection

TTP1 = Tangent to Transition Point-1

CTP1 = Curve Tangent Point-1

CTP2 = Curve Tangent Point-2

TTP2 = Tangent to Transition Point-2

BVC = Beginning of Vertical Curve

EVC = End of Vertical Curve

IP = Intersection Point

RL = Rail Level

G = Gradient

EX = Existing Rail Level

BL = Bed Level

REFERENCE DRAWINGS				
DRAWING TITLE		DRAWING NO.		
DRG.NO.DY.CE(C)KOTA				
H.Q.DR.NO. - CPM/KOT/AT				
NAME OF THE OFFICER & DESIGNATION		SIGNATURE		
(VIJAY PANDEY) CPM/WCR/KOTA				
(GAURAV MISHRA) DY.CE/COTA				
(RAJESH YADAV) AXEN/KOTA				
(P.K. KANWA) SSE/WCR/KOTA				
(S. MEENA) JE/DRG/KOTA				
ISSUE RECORD				
No.	DATE	REV	PREPARED BY	CHECKED BY
1	01-07-2025	R0	SIVA THUTHA	RAMMOHAN
				RAMAKRISHNA

CLIENT:

WEST CENTRAL RAILWAY
KOTA DIVISION

PROJECT : CONDUCTING FINAL LOCATION SURVEY (FLS) INCLUDING PREPARATION OF DPCR, GEOTECHNICAL INVESTIGATION, SIGNALING WORK, ELECTRICAL WORKS AND OPERATING/TRAFFIC WORK FOR CONSTRUCTION OF ELECTRIFIED NEW LINE/DOUBLING/THIRD LINE/4TH LINE. FLY OVER WORK/BYPASS LINE ETC.

TITLE : DETAILED PLAN AND L SECTION MATHURA-NAGDA -3RD LINE
1225+000.000 TO 1230+000.000

CONSULTANT :

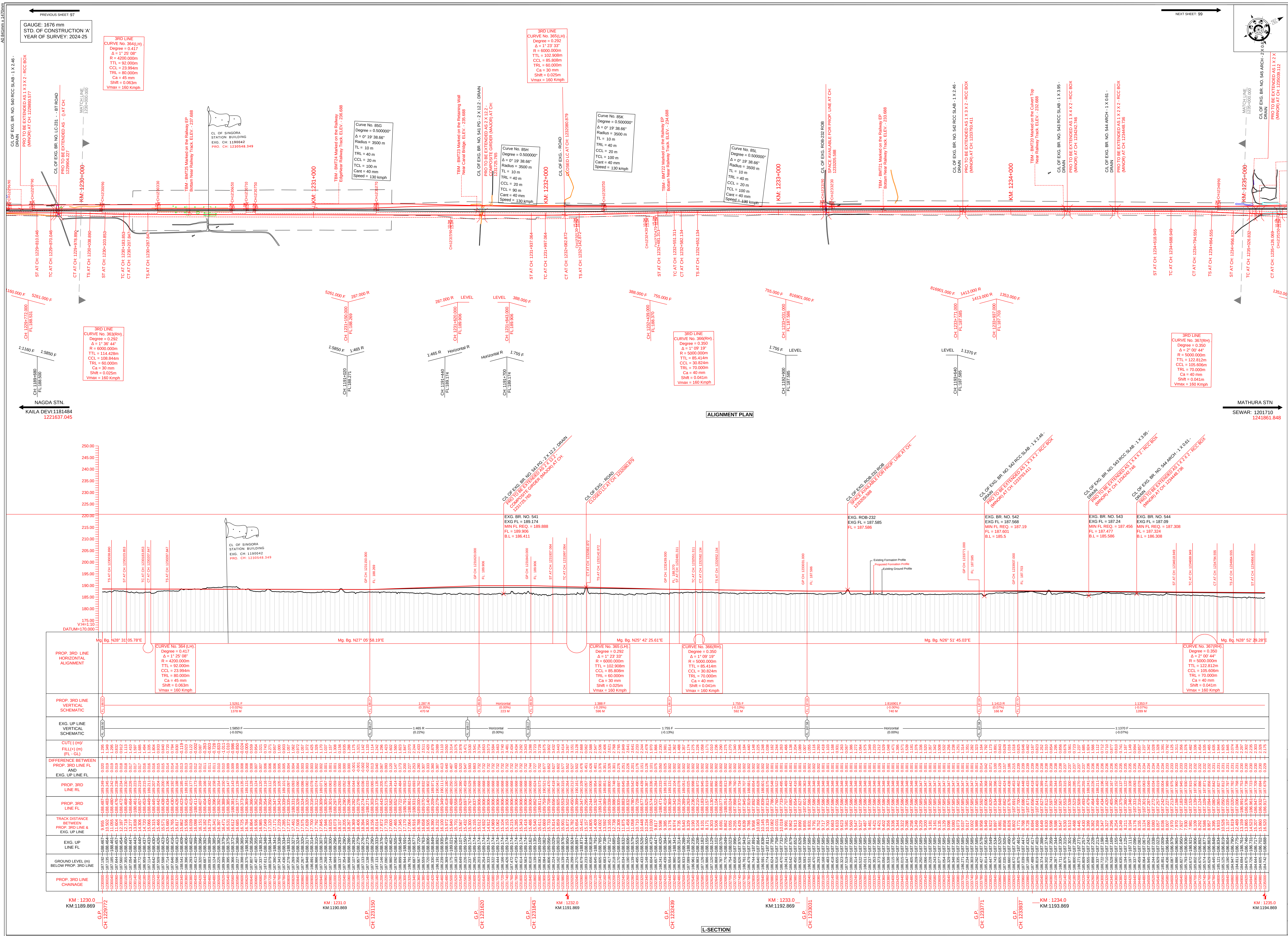
aarvee engineering consultants limited.

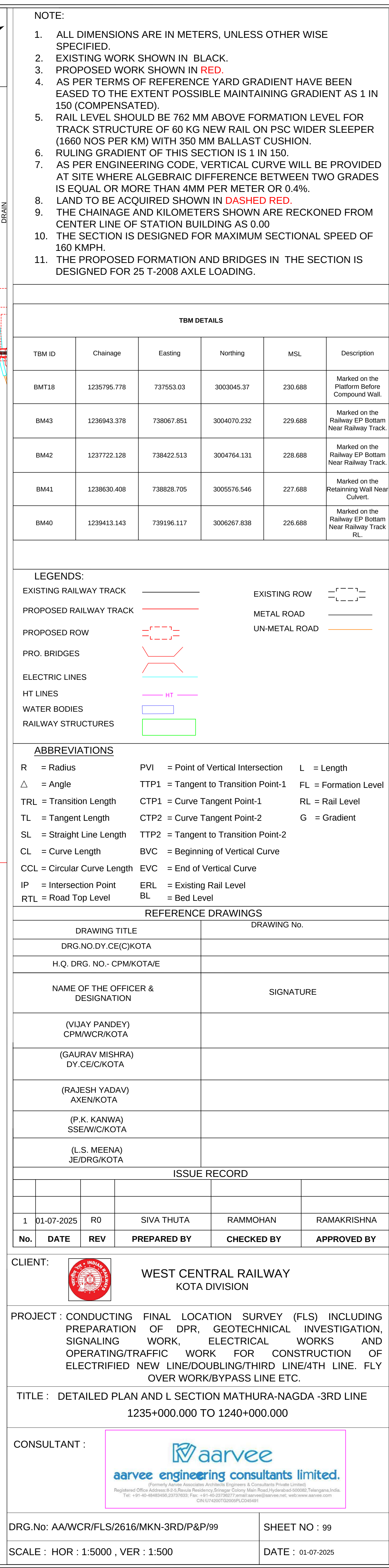
DRG.No: AA/WCR/FLS/2616/MKN-3RD/P&P/97

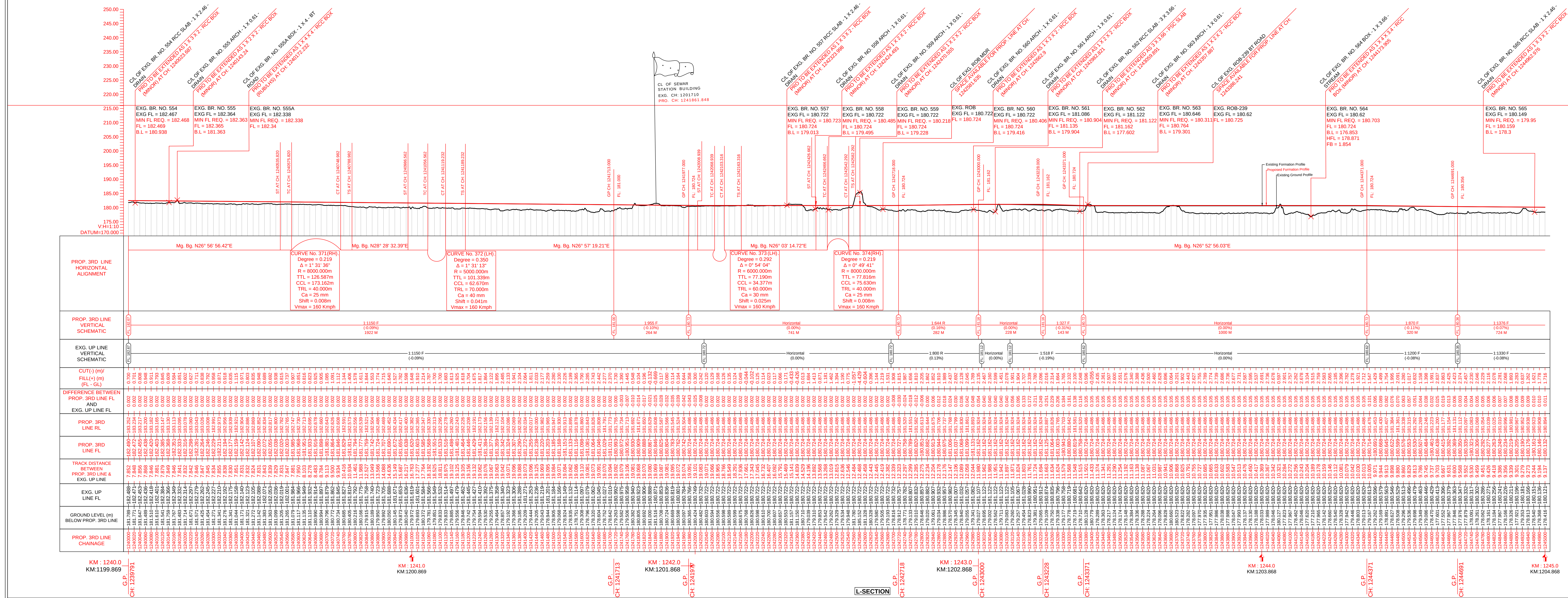
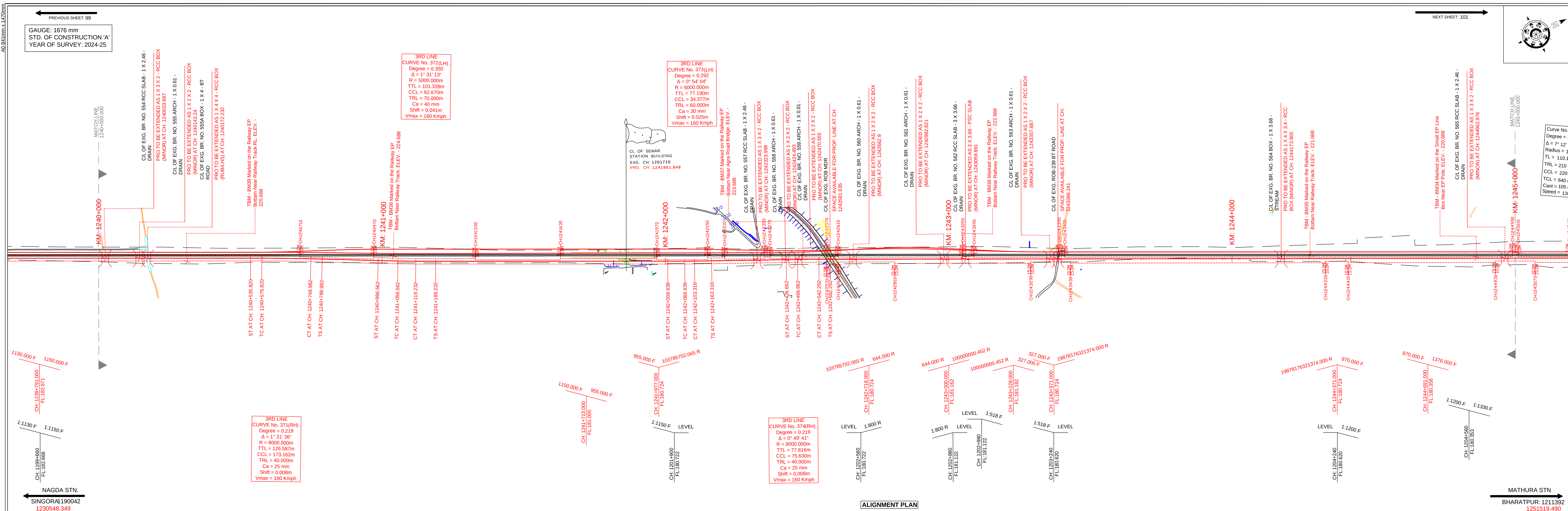
SHEET NO : 97

SCALE : HOR : 1:5000 , VER : 1:500

DATE : 01-07-2025







- NOTE:
1. ALL DIMENSIONS ARE IN METERS, UNLESS OTHERWISE SPECIFIED.
 2. EXISTING WORK SHOWN IN BLACK.
 3. PROPOSED WORK SHOWN IN RED.
 4. AS PER TERMS OF REFERENCE YARD GRADIENT HAVE BEEN EASED TO THE EXTENT POSSIBLE MAINTAINING GRADIENT AS 1 IN 150 (COMPENSATED).
 5. RAIL LEVEL SHOULD BE 762 MM ABOVE FORMATION LEVEL FOR TRACK STRUCTURE OF 1500 MM RAIL ON 950 WIDER SLEEPER (1680 NOS PER KM) WITH 350 MM BALLAST CUSHION.
 6. RULING GRADIENT OF THIS SECTION IS 1 IN 150.
 7. AS PER ENGINEERING CODE, VERTICAL CURVE WILL BE PROVIDED AT SITE WHERE ALGEBRAIC DIFFERENCE BETWEEN TWO GRADES IS EQUAL OR MORE THAN 4MM PER METER OR 0.4%.
 8. LAND TO BE ACQUIRED SHOWN IN DASHED RED.
 9. THE CHAINAGE AND KILOMETERS SHOWN ARE RECKONED FROM CENTER LINE OF SECTION AS 0+00.
 10. THE SECTION IS DESIGNED FOR MAXIMUM SECTIONAL SPEED OF 160 KMPH.
 11. THE PROPOSED FORMATION AND BRIDGES IN THE SECTION IS DESIGNED FOR 25 T-2008 AXLE LOADING.

TBM DETAILS					
TBM ID	Chainage	Easting	Northing	MSL	Description
BM39	1240303.964	739613.243	3007602.984	225.688	Marked on the Railway EP Bottom Near Railway Track RL
BM38	1241039.759	739926.56	3007721.558	224.688	Marked on the Railway EP Bottom Near Railway Track.
BM37	1242497.095	740581.469	3009011.574	223.688	Marked on the Railway EP Bottom Near Agra Road Bridge.
BM36	1243048.268	740837.486	3009511.094	222.688	Marked on the Railway EP Bottom Near Railway Track.
BM35	1244275.677	741391.796	3010606.208	221.688	Marked on the Railway EP Bottom Near Railway Track.
BM34	1244862.511	741663.397	3011126.454	220.688	Marked on the Small EP Line Box Near EP Pole.

LEGENDS:

EXISTING RAILWAY TRACK		EXISTING ROAD	
PROPOSED RAILWAY TRACK		METAL ROAD	
PROPOSED ROW		UN-METAL ROAD	
OVER-BRIDGES			
ELECTRIC LINES			
TELEPHONE LINES			
NATURAL BODIES			
RAILWAY STRUCTURES			

ABBREVIATIONS

= Radius	PVI = Point of Vertical Intersection	L = Length
= Angle	TP1 = Tangent to Transition Point-1	FL = Formation Level
L = Transition Length	CTP1 = Curve Tangent Point-1	RL = Rail Level
= Tangent Length	CTP2 = Curve Tangent Point-2	G = Gradient
= Straight Line Length	TP2 = Tangent to Transition Point-2	
= Curve Length	BVC = Beginning of Vertical Curve	
CL = Circular Curve Length	EVC = End of Vertical Curve	
= Intersection Point	ERL = Existing Rail Level	
TL = Road Top Level	BL = Bed Level	

REFERENCE DRAWINGS

DRAWING TITLE				DRAWING NO.			
DRG.NO.DY.CE/C/KOTA							
H.Q. DRG. NO.-CPM/KOTA/E							
NAME OF THE OFFICER & DESIGNATION				SIGNATURE			
(VIJAY PANDEY) CPM/WCR/KOTA							
(GAURAV MISHRA) DY.CE/C/KOTA							
(RAJESH YADAV) AXEN/KOTA							
(P.K. KANWA) SSE/W/C/KOTA							
(L.S. MEENA) JE/DRG/KOTA							
ISSUE RECORD							
01-07-2025	R0	SIVA THUTA	RAMMOHAN	RAMAKRISHNA			
DATE	REV	PREPARED BY	CHECKED BY	APPROVED BY			

ENT:

WEST CENTRAL RAILWAY
KOTA DIVISION

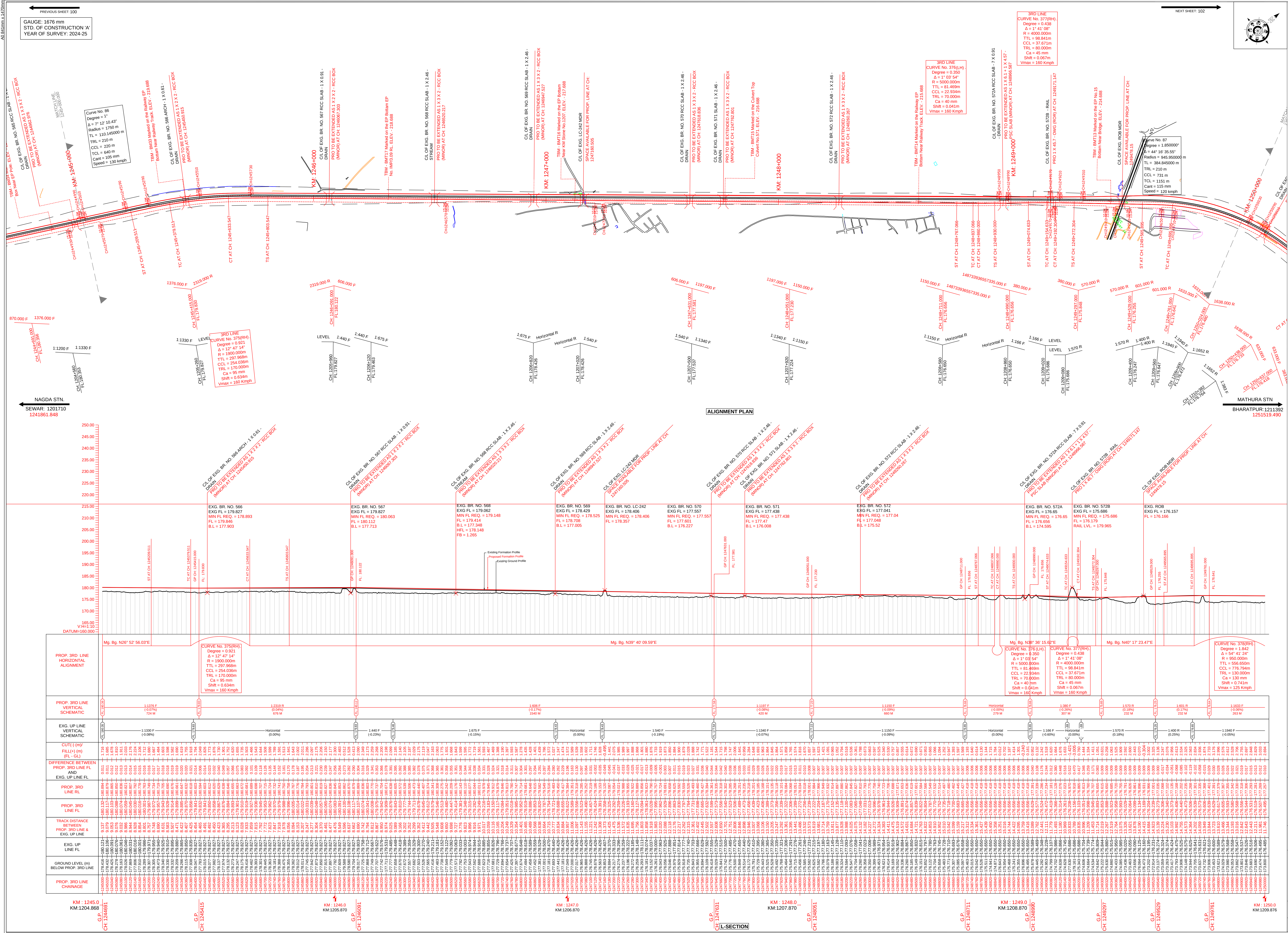
JECT: CONDUCTING FINAL LOCATION SURVEY (FLS) INCLUDING PREPARATION OF DPR, GEOTECHNICAL INVESTIGATION, SIGNALING WORK, ELECTRICAL WORKS AND OPERATING/TRAFFIC WORK FOR CONSTRUCTION OF ELECTRIFIED NEW LINE/DOUBLING/THIRD LINE/4TH LINE. FLY OVER WORK/BYPASS LINE ETC.

LE : DETAILED PLAN AND L SECTION MATHURA-NAGDA -3RD LINE
1240+000.000 TO 1245+000.000

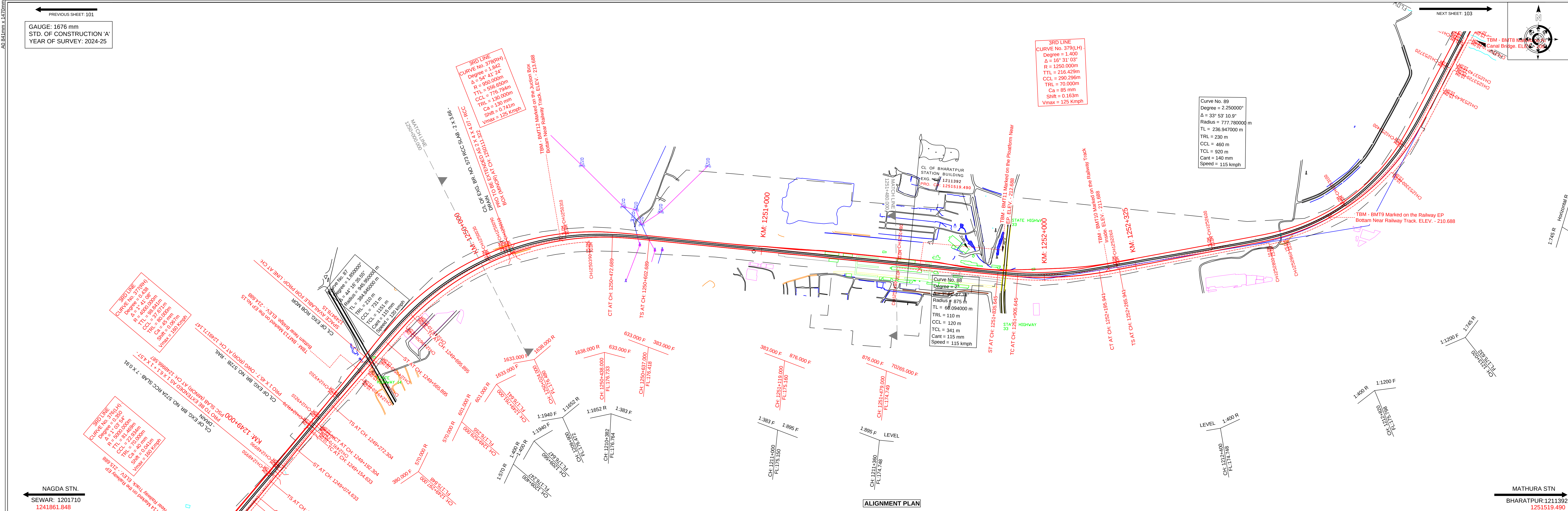
CONSULTANT :



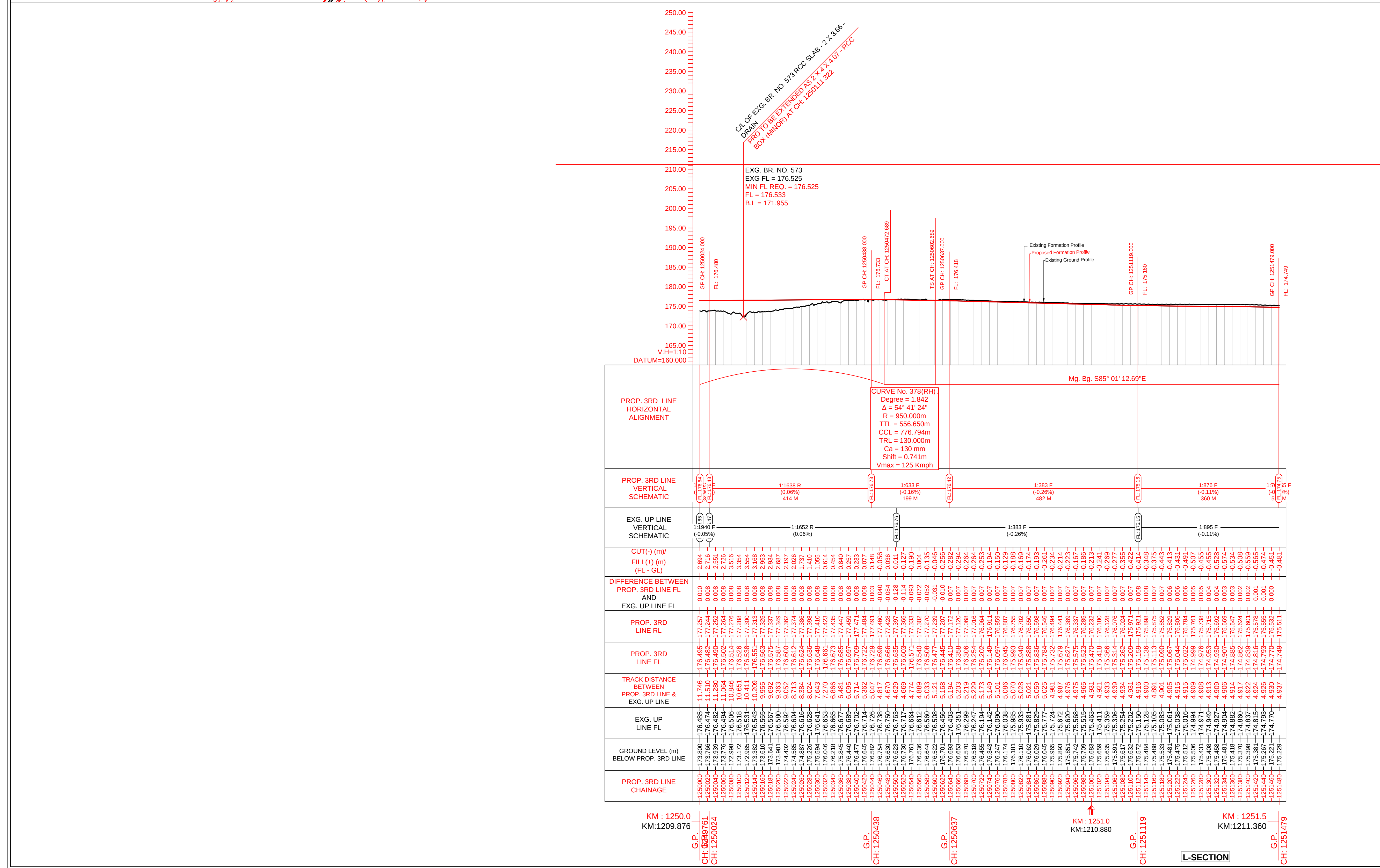
No: AA/WCR/FLS/2616/MKN-3RD/P&P/100 SHEET NO : 100



NOTE:					
1.	ALL DIMENSIONS ARE IN METERS, UNLESS OTHERWISE SPECIFIED.				
2.	EXISTING WORK SHOWN IN BLACK.				
3.	PROPOSED WORK SHOWN IN RED.				
4.	AS PER TERMS OF REFERENCE YARD GRADIENT HAVE BEEN EASED TO THE EXTENT POSSIBLE MAINTAINING GRADIENT AS 1 IN 150 (COMPENSATED).				
5.	RAIL LEVEL SHOULD BE 762 MM ABOVE FORMATION LEVEL FOR TRACK STRUCTURE OF 60 KG NEW RAIL ON PSC WIDER SLEEPER (1660 NOS PER KM) WITH 350 MM BALLAST CUSHION.				
6.	RULING GRADIENT OF THIS SECTION IS 1 IN 150.				
7.	AS PER ENGINEERING CODE, VERTICAL CURVE WILL BE PROVIDED AT SITE WHERE ALGEBRAIC DIFFERENCE BETWEEN TWO GRADES IS EQUAL OR MORE THAN 4MM PER METER OR 0.4%.				
8.	LAND TO BE ACQUIRED SHOWN IN DASHED RED.				
9.	THE CHAINAGE AND KILOMETERS SHOWN ARE RECKONED FROM CENTER LINE OF STATION BUILDING AS 0.00				
10.	THE SECTION IS DESIGNED FOR MAXIMUM SECTIONAL SPEED OF 160 KMPH.				
11.	THE PROPOSED FORMATION AND BRIDGES IN THE SECTION IS DESIGNED FOR 25 T-2008 AXLE LOADING.				
TBM DETAILS					
TBM ID	Chainage	Easting	Northing	MSL	Description
BM33	1245449.193	741926.144	3011652.189	219.688	Marked on the Railway EP Bottom Near Railway Track.
BMT17	1246321.927	742473.669	3012333.496	218.688	Marked on the EP Bottom EP No. BMP/3.09 RL.
BMT16	1247124.371	742998.526	3012949.004	217.688	Marked on the EP Bottom Near KM Stone No.1207.
BMT15	1247769.581	743396.984	3013448.481	216.688	Marked on the Culvert Top Culvert No.571.
BMT14	1248603.097	743930.08	3014089.23	215.688	Marked on the Railway EP Bottom Near Railway Track.
BMT13	1249429.84	744453.521	3014729.263	214.688	Marked on the EP No.15 Bottom Near Bridge.
LEGENDS:					
EXISTING RAILWAY TRACK			EXISTING ROW		
PROPOSED RAILWAY TRACK			METAL ROAD		
PROPOSED ROW			UN-METAL ROAD		
PRO. BRIDGES					
ELECTRIC LINES					
HT LINES					
WATER BODIES					
RAILWAY STRUCTURES					
ABBREVIATIONS					
R = Radius	PVI = Point of Vertical Intersection		L = Length		
Δ = Angle	TTP1 = Tangent to Transition Point-1		FL = Formation Level		
TRL = Transition Length	CTP1 = Curve Tangent Point-1		RL = Rail Level		
TL = Tangent Length	CTP2 = Curve Tangent Point-2		G = Gradient		
SL = Straight Line Length	TTP2 = Tangent to Transition Point-2				
CL = Curve Length	BVC = Beginning of Vertical Curve				
CCL = Circular Curve Length	EVC = End of Vertical Curve				
IP = Intersection Point	ERL = Existing Rail Level				
RTL = Road Top Level	BL = Bed Level				
REFERENCE DRAWINGS					
DRAWING TITLE			DRAWING No.		
DRG.No.DY.CE(C)/KOTA					
H.Q. DRG. NO. - CPM/KOTA/E					
NAME OF THE OFFICER & DESIGNATION			SIGNATURE		
(VIJAY PANDEY) CPM/WCR/KOTA					
(GAURAV MISHRA) DY.CE/CIKOTA					
(RAJESH YADAV) AXEN/KOTA					
(P.K. KANWA) SSE/WC/KOTA					
(L.S. MEENA) JE/DRG/KOTA					
ISSUE RECORD					
1	01-07-2025	R0	SIVA THUTA	RAMMOHAN	RAMAKRISHNA
No.	DATE	REV	PREPARED BY	CHECKED BY	APPROVED BY
CLIENT:					
 WEST CENTRAL RAILWAY KOTA DIVISION					
PROJECT : CONDUCTING FINAL LOCATION SURVEY (FLS) INCLUDING PREPARATION OF DPR, GEOTECHNICAL INVESTIGATION, SIGNALING WORK, ELECTRICAL WORKS AND OPERATING/TRAFFIC WORK FOR CONSTRUCTION OF ELECTRIFIED NEW LINE/DOUBLING/THIRD LINE/4TH LINE. FLY OVER WORK/BYPASS LINE ETC.					
TITLE : DETAILED PLAN AND L SECTION MATHURA-NAGDA -3RD LINE 1245+000.000 TO 1250+000.000					
CONSULTANT :					
 aarvee engineering consultants limited. Registered Office: 501, 5th Floor, New Axis Centre, Regd. Office: 208/2, 2nd Floor, New Axis Centre, Main Road, Indraprastha-110002, Patparganj, India. Tel: +91-8860020277/2770377 Fax: +91-882250577 Email: info@aarvee.com, info@aarvee.co.in GSA: 14102102000					



ALIGNMENT PLAN



L-SECTION

- NOTE:
- ALL DIMENSIONS ARE IN METERS, UNLESS OTHERWISE SPECIFIED.
 - EXISTING WORK SHOWN IN BLACK.
 - PROPOSED WORK SHOWN IN RED.
 - AS PER TERMS OF REFERENCE YARD GRADIENT HAVE BEEN EASED TO THE EXTENT POSSIBLE MAINTAINING GRADIENT AS 1 IN 150 (COMPENSATED).
 - RAIL LEVEL SHOULD BE 762 MM ABOVE FORMATION LEVEL FOR TRACK STRUCTURE OF 60 KG NEW RAIL ON PSC WIDER SLEEPER (1660 NOS PER KM) WITH 350 MM BALLAST CUSHION.
 - RULING GRADIENT OF THIS SECTION IS 1 IN 150.
 - AS PER ENGINEERING CODE, VERTICAL CURVE WILL BE PROVIDED AT SITE WHERE ALGEBRAIC DIFFERENCE BETWEEN TWO GRADES IS EQUAL OR MORE THAN 4MM PER METER OR 0.4%.
 - LAND TO BE ACQUIRED SHOWN IN DASHED RED.
 - THE CHAINAGE AND KILOMETERS SHOWN ARE RECKONED FROM CENTER LINE OF STATION BUILDING AS 0.00
 - THE SECTION IS DESIGNED FOR MAXIMUM SECTIONAL SPEED OF 160 KMPH.
 - THE PROPOSED FORMATION AND BRIDGES IN THE SECTION IS DESIGNED FOR 25 T-2008 AXLE LOADING.

TBM DETAILS					
TBM ID	Chainage	Existing	Northing	MSL	Description
BMT12	1250290.573	745152.137	3015202.719	213.688	Marked on the Junction Box Bottom Near Railway Track

LEGENDS:	
EXISTING RAILWAY TRACK	EXISTING ROW
PROPOSED RAILWAY TRACK	METAL ROAD
PROPOSED ROW	UN-METAL ROAD
PRO. BRIDGES	
ELECTRIC LINES	
HT LINES	
WATER BODIES	
RAILWAY STRUCTURES	

ABBREVIATIONS	
R = Radius	PVI = Point of Vertical Intersection
Δ = Angle	TTP1 = Tangent to Transition Point-1
TRL = Transition Length	CTP1 = Curve Tangent Point-1
TL = Tangent Length	CTP2 = Curve Tangent Point-2
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CCL = Circular Curve Length	EVC = End of Vertical Curve
IP = Intersection Point	ERL = Existing Rail Level
RTL = Road Top Level	BL = Bed Level

REFERENCE DRAWINGS	
DRAWING TITLE	DRAWING No.
DRG.NO.DY.CE(C)KOTA	
H.Q. DRG. NO. - CPM/KOTA/E	
NAME OF THE OFFICER & DESIGNATION	SIGNATURE
(VIJAY PANDEY) CPM/WCR/KOTA	
(GAURAV MISHRA) DY.CE/C/KOTA	
(RAJESH YADAV) AXEN/KOTA	
(P.K. KANWA) SSE/WIC/KOTA	
(L.S. MEENA) JE/DRG/KOTA	

ISSUE RECORD				
No.	DATE	REV	PREPARED BY	CHECKED BY
1	01-07-2025	R0	SIVA THUTA	RAMMOHAN
				RAMAKRISHNA

CLIENT:	WEST CENTRAL RAILWAY KOTA DIVISION
---------	---------------------------------------

PROJECT : CONDUCTING FINAL LOCATION SURVEY (FLS) INCLUDING PREPARATION OF DPR, GEOTECHNICAL INVESTIGATION, SIGNALING WORK, ELECTRICAL WORKS AND OPERATING/TRAFFIC WORK FOR CONSTRUCTION OF ELECTRIFIED NEW LINE/DOUBLING/THIRD LINE/4TH LINE. FLY OVER WORK/BYPASS LINE ETC.

TITLE : DETAILED PLAN AND L SECTION MATHURA-NAGDA -3RD LINE
1250+000.000 TO 1252+325.056

CONSULTANT :	aarvee aarvee engineering consultants limited.
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DRG.No: AA/WCR/FLS/2616/MKN-3RD/P&P/102	SHEET NO : 102
SCALE : HOR : 1:5000 , VER : 1:500	DATE : 01-07-2025